Plain-language explanation: heart-s1-c1

AI-assisted drafts for review. Nothing here has been verified by a person.

How to read this. Explanations are built up but read down: the summary first, then each part from the whole model down, in plain words, each sentence followed by the claims it rests on (C01, C02, ...; every cited claim was computed on the model and checked). Numbers are changes in predicted risk for a typical patient. The technical record (original hypotheses, formulas, evidence tables with the test check, ranges and n) is at the end. The text describes what the model does, not the world; whether it is plausible is for the reviewer to judge.

What changed in v5e (figures). v5e: every diagram is a numbered figure with a caption (list of figures after the contents) and the text refers to it; the whole-network figure has its own landscape page, scaled to the page, with a short caption and its description in the text; an input used by several parts is drawn under each of them (same input, repeated for readability, dotted outline); and an appendix figure shows the trained network before any explanation (no parts, no splits).

What changed in v5c (tables and diagrams). v5c: the evidence tables are landscape, compact and shaded by row, with the header repeated on every page (longtable and pdflscape, used in this review PDF only; the thesis preamble has neither). In the diagrams each input is drawn once, in a shared row along the bottom (the explanation's split copies of an input are listed in the technical detail instead), every cluster has one output rectangle with its plain name, the inputs that feed the output directly meet in a drawn 'sum', and a one-line legend explains the edges.

What changed in v5b (diagrams). v5b uses one grouping everywhere, the annotation hierarchy: the summary's diagram is the same picture as the whole model's section, and a part that is itself a weighted sum (blood pressure with cholesterol) is a part of its own, described as one that adds up, so the text counts three parts throughout; the lint checks every 'N parts' against the hierarchy. A full-structure diagram follows the summary: every node and connection, nested by part, with the inputs that feed the output directly in their own cluster and the unused input columns listed. Each section from the top down adds its own region of that diagram, with stubs for connections from or to other parts; each connection is drawn in exactly one section. The gate's text now says that it lowers the risk for the patients it acts on, and that more vessels and higher cholesterol take that lowering away; its vessel statements are one per group.

What changed in v5 (read down). v5 reads the explanation down. The summary (one line, a diagram, one paragraph per block) is as in v4, except that the paragraph on what adds up is split into protective items, risk-raising items and rare results, and an input that also feeds a part is quoted with its direct contribution only (the rest is in that part's paragraph). Then the document walks the annotation hierarchy from the whole model down: for each part a plain hypothesis (what it does for the prediction), plain evidence (what was checked and what it showed; for a part made of others, what they are and how they combine), a diagram one level down and where to read on. The original hypotheses, closed forms and evidence tables are in 'Technical detail' at the end. Statements about structure (what a part is made of, a copy of another part, a formula that reproduces it) are now claims of their own, checked on the graph and the data.

v4. v4 follows the structure of the explanation. The summary opens with one line and a diagram at the same grouping, then gives one paragraph per top-level block of the annotation hierarchy: one for everything that adds up independently (the inputs that feed the model score directly, and parts that are a weighted sum), and one for each part that combines its inputs in another way: what it combines, how, and for whom. 'Parts in detail' follows the hierarchy below that, with a diagram one level down for each part that contains others. The prose no longer says 'clearly', 'slightly', 'consistently' or 'on the training data only': the numbers carry it, and the test check, the ranges and n stay in the evidence tables. Only genuinely small groups are marked, with their n. The lint rejects the confidence words and checks the block structure.

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1 heart-s1-c1

Genome `a2df22ce-983d-47b6-bbd6-3176d38ed5ff`, explanation `492866e6-a73b-4465-92a8-ecad82a2adcc`. Split: train primary (n=242), test as a check (n=61). Claims: 47, verified 47, not supported 0.
AI-assisted draft for review (unverified)

Reference (typical) patient: the typical patient: the median training patient for every number and the most common training category for every category (not a real patient). age: 56 years; ca: 0 vessels; chol: 240 mg/dL (6.2 mmol/L); cp: no chest pain (asymptomatic); exang: no chest pain on exercise; fbs: fasting blood sugar of 120 mg/dL or less; oldpeak: 0.8 mm; restecg: a normal resting ECG; sex: male; slope: an upsloping ST segment; thal: a normal thallium scan; thalach: 152 beats per minute; trestbps: 130 mm Hg. Predicted risk of heart disease: 18%.

1.1 Summary

For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 18%; it adds up most of its inputs independently, and five parts add to that sum: two add more for each extra vessel coloured, one lowers the risk for typical angina, one lowers it for non-anginal chest pain, one acts only on the 42 of 242 training patients with a normal resting ECG and marked ST depression, and one lowers it for a higher maximum heart rate, with nothing until about 145 beats per minute, then steadily. [C29, C30, C40, C31, C43, C38, C46] Figure 1 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards heart disease; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, normal ECG has a positive connection into 'ST depression with a normal ECG', a part that raises the predicted risk of heart disease, so normal ECG raises it.

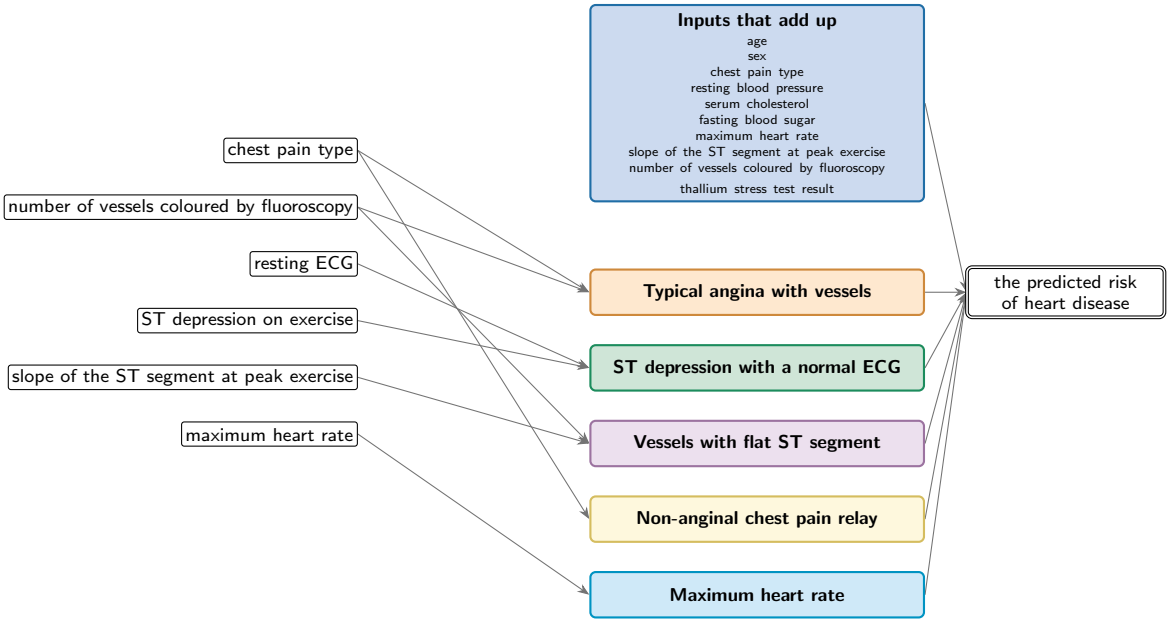


Figure 1: The model at the level of the summary: one box per part.

Inputs that add up. *Lowering the risk.* These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 18% to below 1% and fasting blood sugar above 120 mg/dL to about 8%; through this sum alone, compared with no chest pain symptoms, typical angina or non-anginal chest pain lowers it to about 2% and atypical angina to about 5% (chest pain type also acts through the parts below). [C10, C19, C12, C17]

Raising the risk. A reversible defect (or the undefined code 4) on the thallium scan raises it to about 55%; higher cholesterol (from about 8% at 193 mg/dL to about 45% at 308 mg/dL), higher resting blood pressure (from about 9% at 110 mm Hg to about 35% at 155 mm Hg) and older age (from about 10% at 42 years to about 25% at 66 years) raise it steadily. [C05, C07, C08, C13] Through this sum alone, each extra vessel coloured by fluoroscopy adds about the same (from about 18% at 0 to about 80% at 2 vessels) and a flat ST segment raises it to about 87%, while the maximum heart rate barely changes it; all three also act through the parts below. [C04, C03, C21]
Rare results. Rarer results also count (the model has little data on this, n = 15 and 17): a fixed defect on the thallium scan lowers the risk to about 3%, and through this sum alone a downsloping ST segment raises it to about 32%; the model ignores exercise-induced angina. [C14, C15, C28]

Typical angina with vessels. This part combines chest pain type with the number of vessels coloured: through it, typical angina lowers a typical patient's predicted risk from about 18% to about 2%, and each extra vessel adds about the same, from about 18% at 0 to about 61% at 2 vessels. [C31, C30] Across the whole model, more vessels raise the predicted risk, and even more so with typical angina, each extra vessel adding about the same: going from 0 to 2 vessels moves a typical patient from below 1% to about 55% with typical angina, and from about 18% to about 99% with any other chest pain type. [C32] Across the whole model, typical angina takes a typical patient's predicted risk to below 1%. [C09]

ST depression with a normal ECG. This part combines the resting ECG with ST depression on exercise and acts on 42 of 242 training patients, nearly all with a normal resting ECG, 26 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 98% without it and about 99% with it. [C38] Through it, more ST depression raises the predicted risk only once it rises above about 1.0 mm, for patients with a normal resting ECG, in an uneven way: for a typical such patient from about 13% at 0 mm to about 63% at 3.5 mm. [C37] More ST depression therefore raises the predicted risk, but only if the resting ECG is normal: going from 0 to 2.8 mm moves a typical patient from about 18% to about 55% with a normal resting ECG, in an uneven way, and leaves it at about 18% otherwise. [C36]

Vessels with flat ST segment. This part combines the number of vessels coloured with a flat ST segment: through it, each extra vessel adds about the same, from about 18% at 0 to about 36% at 2 vessels for a typical patient, and a flat ST segment raises the risk from about 18% to about 23%. [C40, C41] Across the whole model, each extra vessel takes a typical patient from about 18% at 0 to about 99% at 2 vessels, about the same for each, flattening out near the top because the predicted risk cannot go much higher, and a flat ST segment takes it to about 91%. [C01, C02]

Non-anginal chest pain relay. This part reads only non-anginal chest pain: through it, non-anginal chest pain lowers a typical patient's predicted risk from about 18% to about 8%. [C43] Across the whole model, non-anginal chest pain takes it to about 1%. [C11]

Maximum heart rate. This part reads only the maximum heart rate: a higher maximum heart rate lowers the predicted risk with nothing until about 145 beats per minute, then steadily, for a typical patient from about 19% at 118 to about 14% at 175 beats per minute. [C45, C46] Across the whole model the same holds, from about 19% at 118 to about 14% at 175 beats per minute. [C20]

The whole network. Figure 2 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, ...); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): male, sugar 120 or less, ST-T abnormal ECG, LVH on ECG, no exercise pain, exercise pain, upsloping ST, thallium code 4, reversible defect.

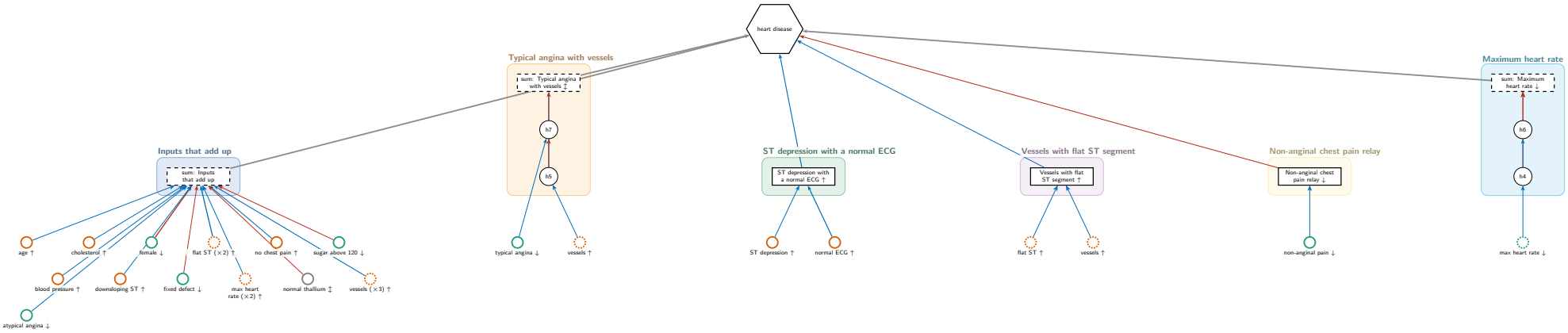


Figure 2: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: patients with typical angina or non-anginal chest pain get the lowest predicted risk of all chest pain types (below 1% and about 1% for a typical patient), below patients with no chest pain symptoms. [C09, C11] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C14] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 18% to about 8%). [C19] Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 18% to about 99%. [C01] Worth checking: ST depression on exercise raises the predicted risk only when the resting ECG is normal. [C36] Worth checking: the model ignores exercise-induced angina. [C28]

1.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

1.2.1 The whole model

Figure 3 shows what feeds this part; Figure 4 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

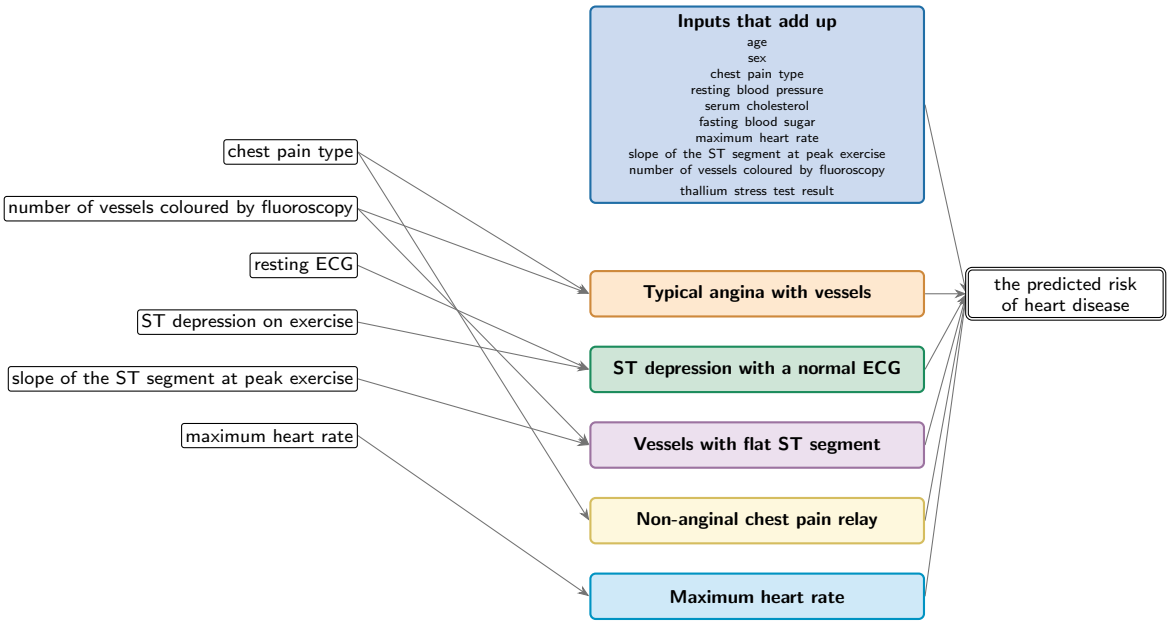


Figure 3: The whole model: what feeds it, one level down.

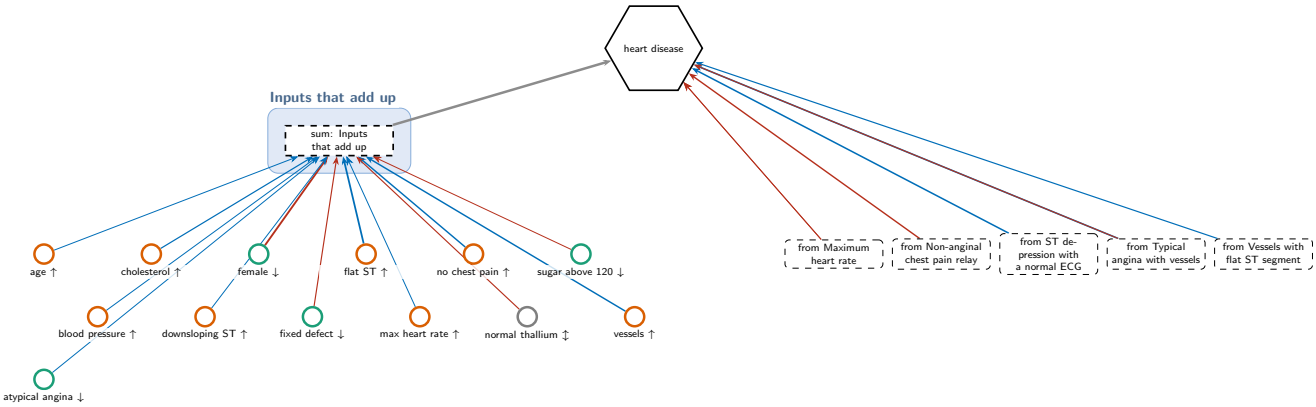


Figure 4: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, chest pain type and sex each move the predicted risk on their own, five parts add to

that sum, and exercise-induced angina is ignored. [C29, C01, C02, C10, C28]

Evidence. It is made up of the inputs that feed the score directly and five parts: vessels with a flat ST segment, the non-anginal chest pain relay, ST depression with a normal ECG, the maximum heart rate part and typical angina with vessels (each explained below); the score is their sum, which was checked on all 242 training patients. [C29] Changing one input at a time across the whole model, each extra vessel adds about the same, taking a typical patient's predicted risk from about 18% at 0 to about 99% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher, against about 80% through the sum alone. [C01, C04] More ST depression raises it in an uneven way across the whole model, from about 18% at 0 mm to about 55% at 2.8 mm; it enters the score only through the part with a normal ECG. [C06] The model ignores exercise-induced angina. [C28]
Evidence plots: Figure 5.

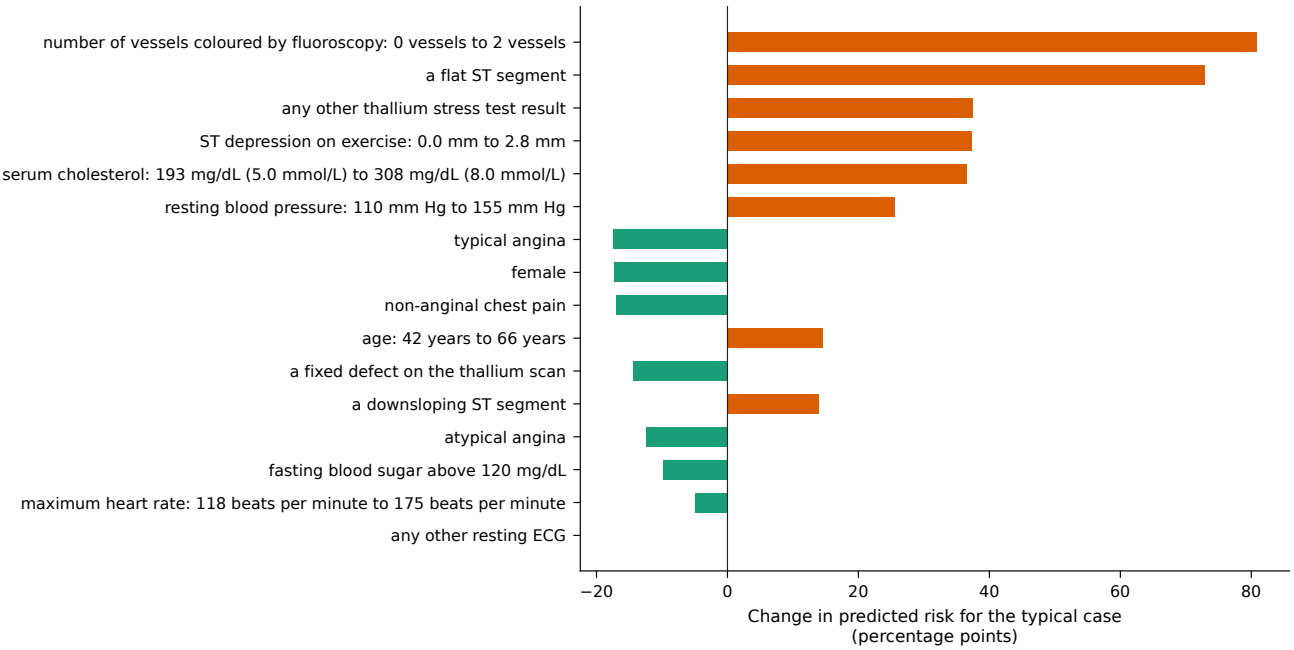


Figure 5: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

Dive deeper: Typical angina with vessels (1.2.2); ST depression with a normal ECG (1.2.3); Vessels with flat ST segment (1.2.4); Non-anginal chest pain relay (1.2.5); Maximum heart rate (1.2.6). Technical detail: T.1.

1.2.2 Typical angina with vessels

Part of: The whole model (1.2.1).
Figure 6 shows what feeds this part; Figure 7 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

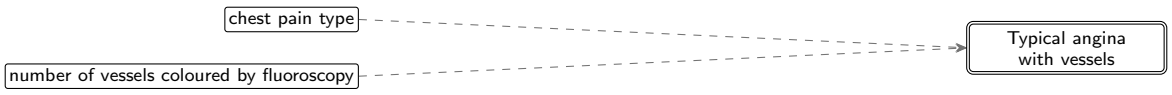


Figure 6: Typical angina with vessels: what feeds it, one level down.

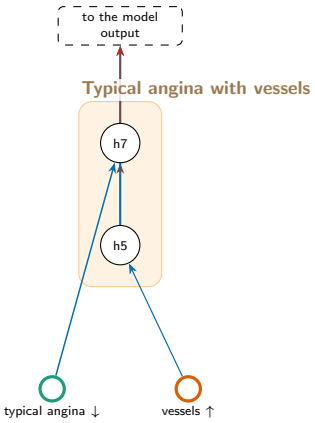


Figure 7: Typical angina with vessels: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for typical angina and adds risk for each extra vessel coloured, about the same for each, with the vessels counting more for patients with typical angina. [C31, C30, C32]

Evidence. Through it, typical angina takes a typical patient's predicted risk from about 18% to about 2%, and each extra vessel adds about the same, from about 18% at 0 to about 61% at 2 vessels. [C31, C30] Across the whole model, going from 0 to 2 vessels moves a typical patient from below 1% to about 55% with typical angina, each extra vessel adding about the same, and from about 18% to about 99% with any other chest pain type. [C32] Its formula reproduces the part exactly on all 242 training patients. [C33]
Evidence plots: Figure 8, Figure 9, Figure 10, Figure 11, Figure 12, Figure 13.
Figure 11 shows this part's own output, unit h5.
Figure 12 shows this part's own output, unit h7.

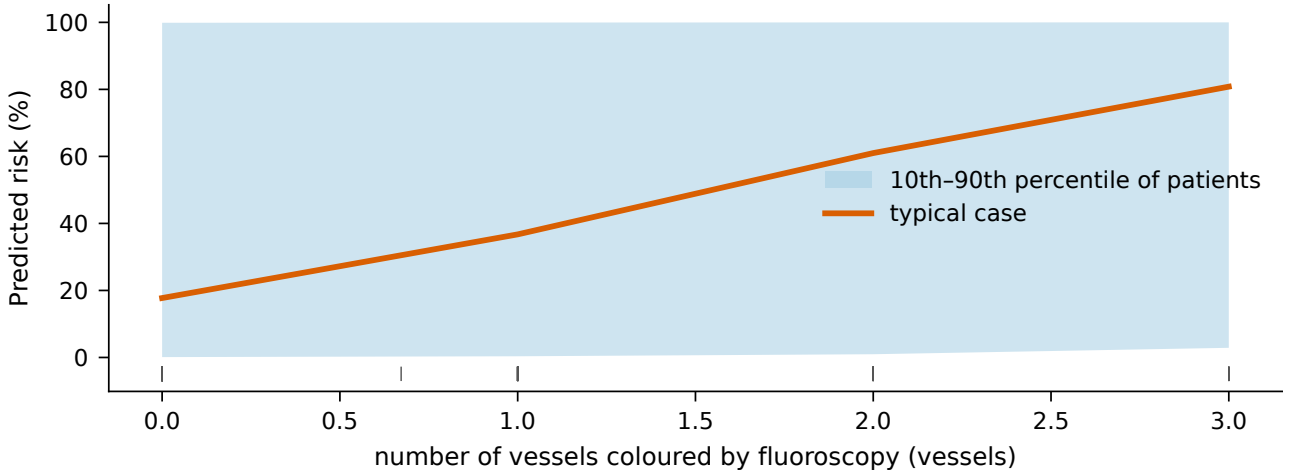


Figure 8: Typical angina with vessels: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

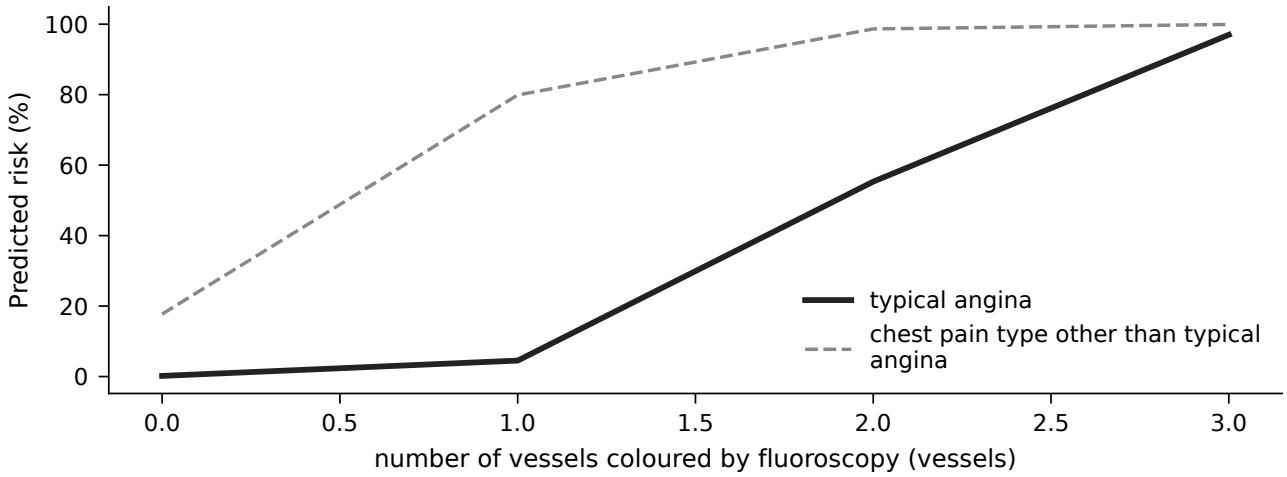


Figure 9: Typical angina with vessels: How the effect differs between groups: one curve per group, for the typical case of each group.

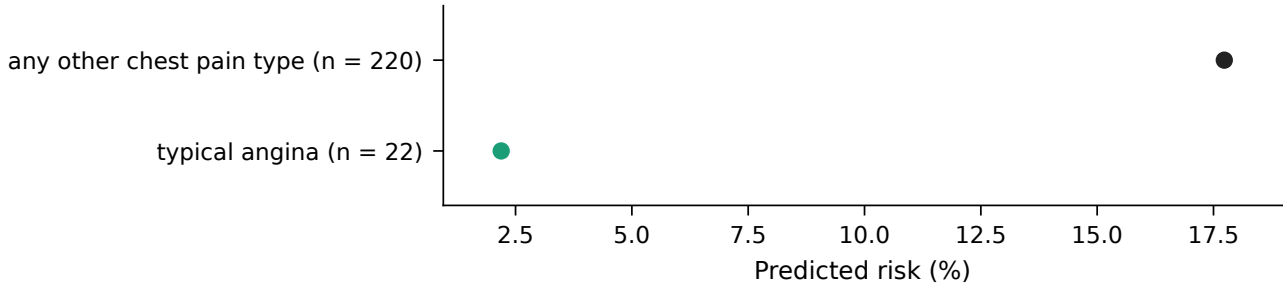


Figure 10: Typical angina with vessels: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

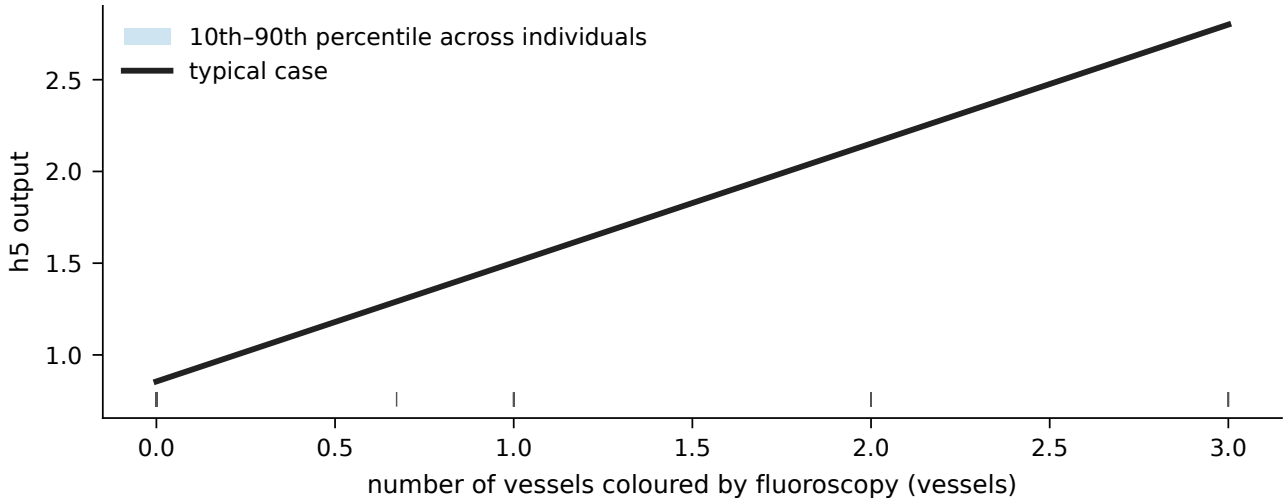


Figure 11: Typical angina with vessels: This part's own output, unit h5 (the output of 'Typical angina with vessels'), on its own scale (its activation), with every other input at the typical case; it is active for 242 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; ticks: the data.

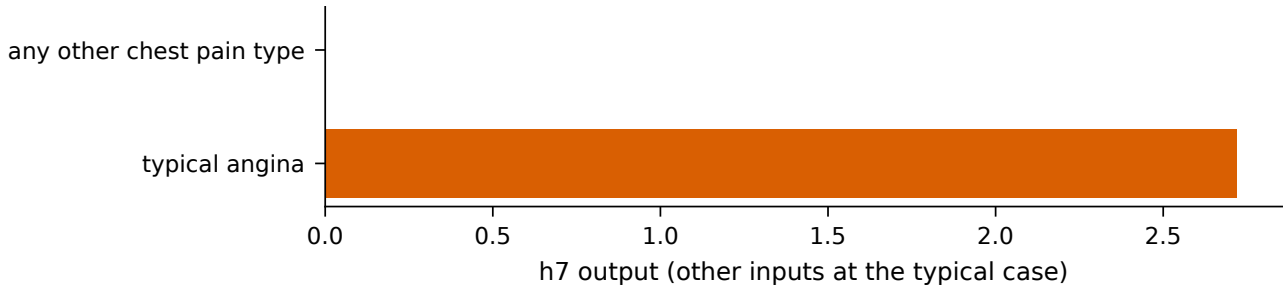


Figure 12: Typical angina with vessels: This part's own output, unit h7 (the output of 'Typical angina with vessels'), on its own scale (its activation), with every other input at the typical case; it is active for 22 of 242. Its output for each combination of the levels it reads.

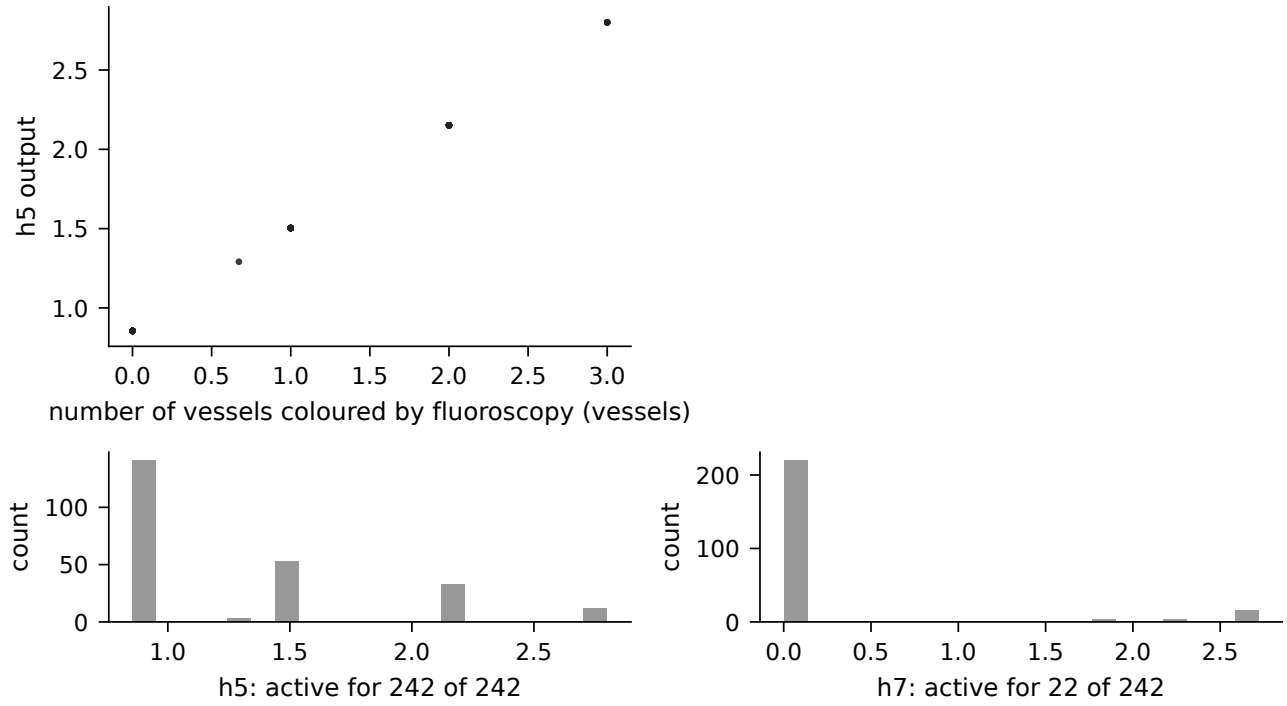


Figure 13: Typical angina with vessels: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

1.2.3 ST depression with a normal ECG

Part of: The whole model (1.2.1).
Figure 14 shows what feeds this part; Figure 15 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

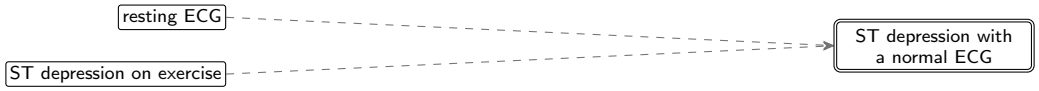


Figure 14: ST depression with a normal ECG: what feeds it, one level down.

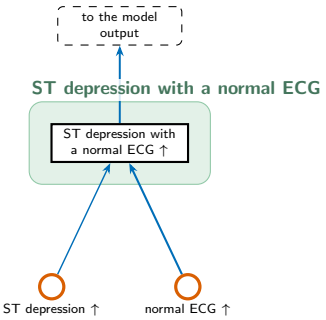


Figure 15: ST depression with a normal ECG: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the predicted risk with ST depression on exercise, in an uneven way, only for patients with a normal resting ECG, once it rises above about 1.0 mm. [C37, C36]

Evidence. It acts on 42 of 242 training patients, nearly all with a normal resting ECG, 26 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 98% without it and about 99% with it. [C38] Through it, more ST depression raises the predicted risk only once it rises above about 1.0 mm, in an uneven way: for a typical patient with a normal resting ECG from about 13% at 0 mm to about 63% at 3.5 mm. [C37] Going from 0 to 2.8 mm of ST depression moves a typical patient from about 18% to about 55% with a normal resting ECG, in an uneven way, and leaves it at about 18% otherwise. [C36] Its formula reproduces the part exactly on all 242 training patients. [C39]
Evidence plots: Figure 16, Figure 17, Figure 18, Figure 19, Figure 20.
Figure 19 shows this part's own output, unit h3: it switches on at about 1.006 mm of ST depression on exercise, and is zero below.

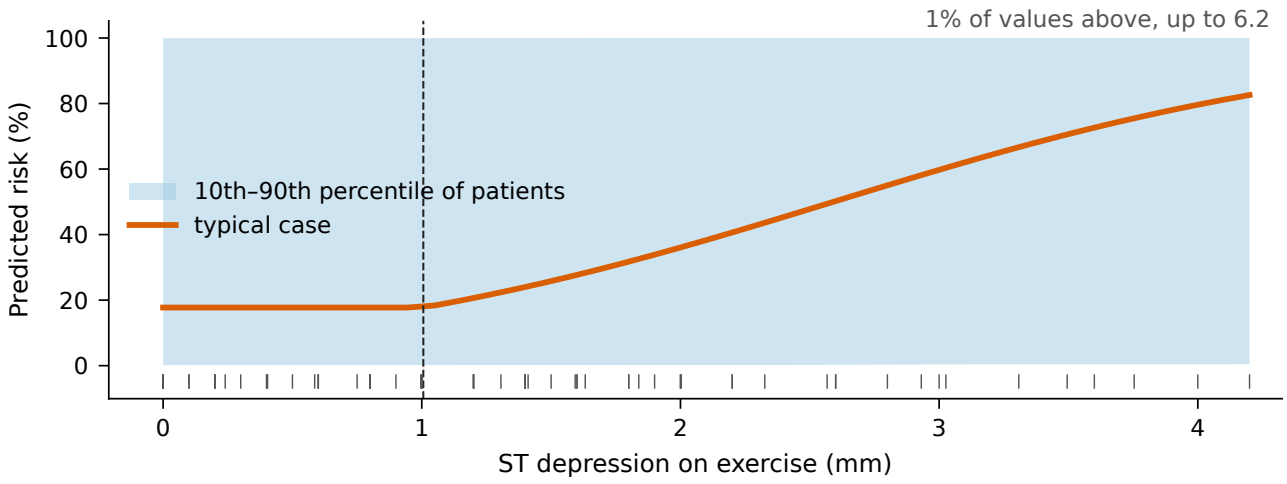


Figure 16: ST depression with a normal ECG: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

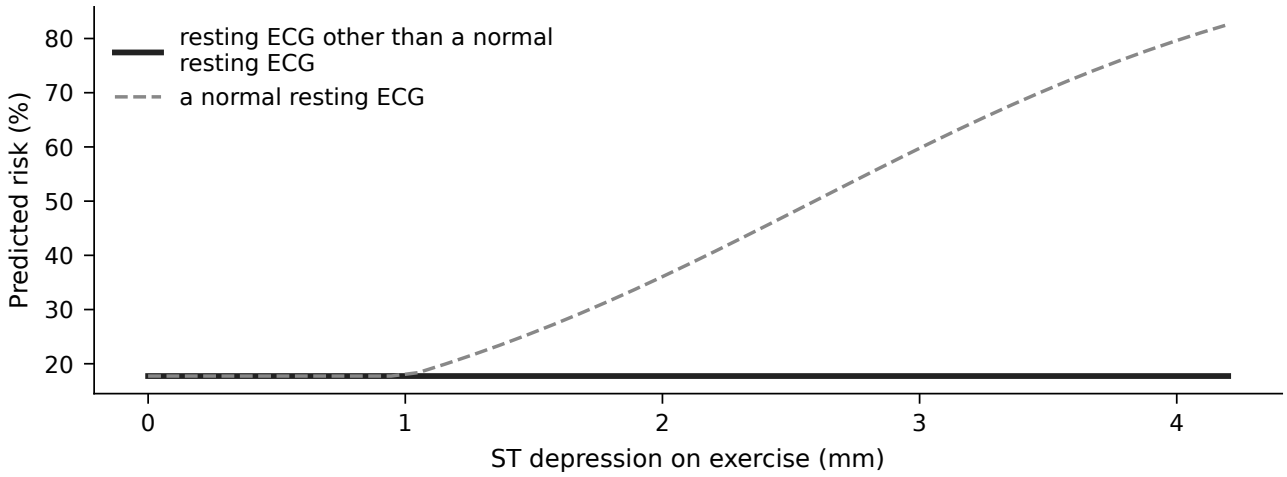


Figure 17: ST depression with a normal ECG: How the effect differs between groups: one curve per group, for the typical case of each group.

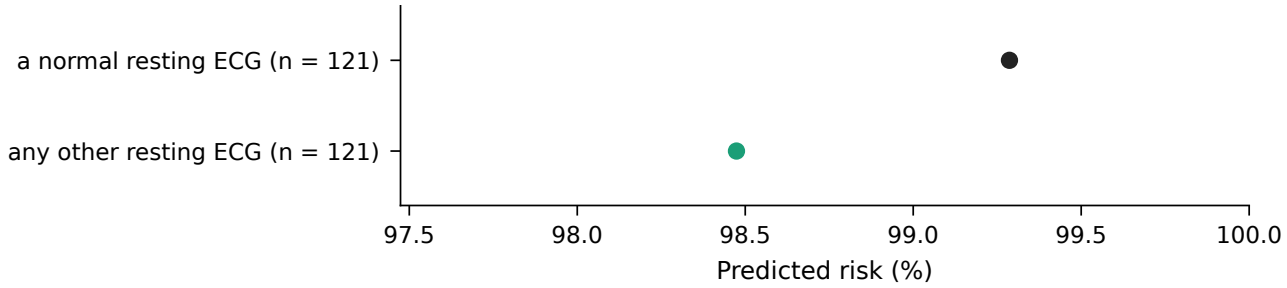


Figure 18: ST depression with a normal ECG: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

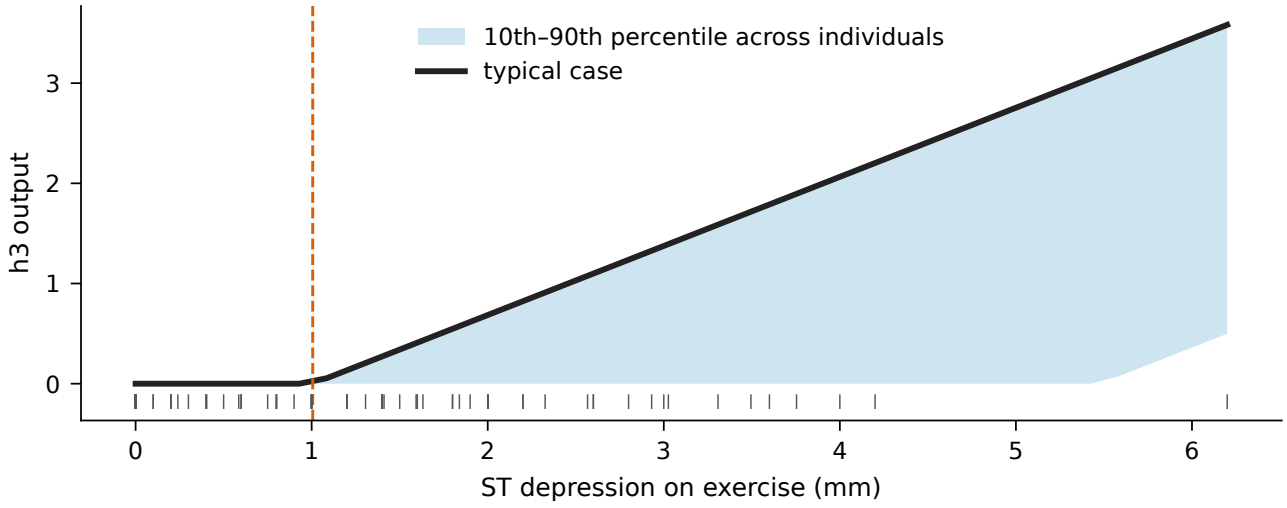


Figure 19: ST depression with a normal ECG: This part's own output, unit h3 (the output of 'ST depression with a normal ECG'), on its own scale (its activation), with every other input at the typical case; it is active for 42 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

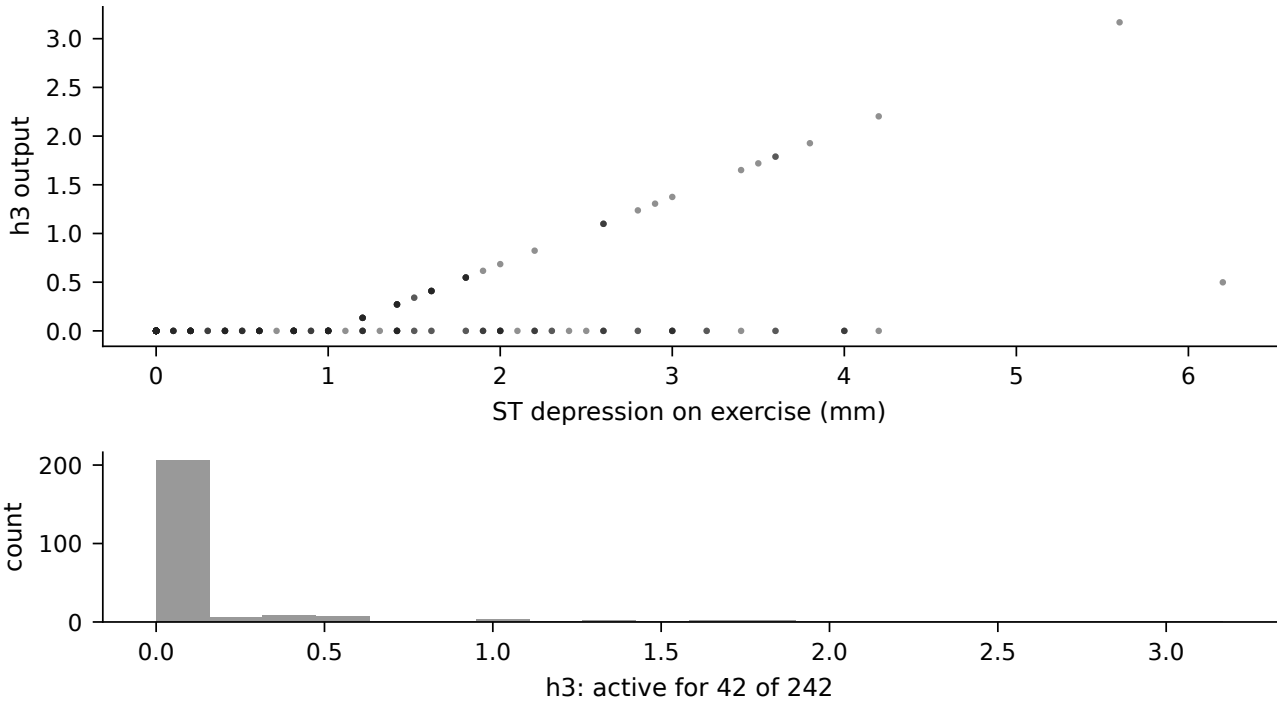


Figure 20: ST depression with a normal ECG: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.3.

1.2.4 Vessels with flat ST segment

Part of: The whole model (1.2.1).
Figure 21 shows what feeds this part; Figure 22 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

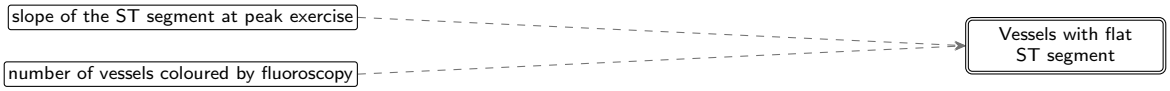


Figure 21: Vessels with flat ST segment: what feeds it, one level down.

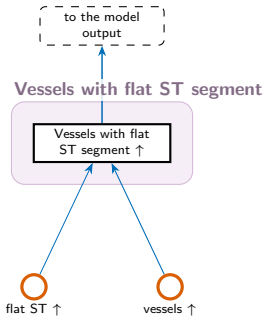


Figure 22: Vessels with flat ST segment: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds risk for each extra vessel coloured, about the same for each, and for a flat ST segment. [C40, C41]

Evidence. Through it, each extra vessel adds about the same, from about 18% at 0 to about 36% at 2 vessels for a typical patient, and a flat ST segment raises the risk from about 18% to about 23%. [C40, C41] Its formula reproduces the part exactly on all 242 training patients. [C42]

Evidence plots: Figure 23, Figure 24, Figure 25.

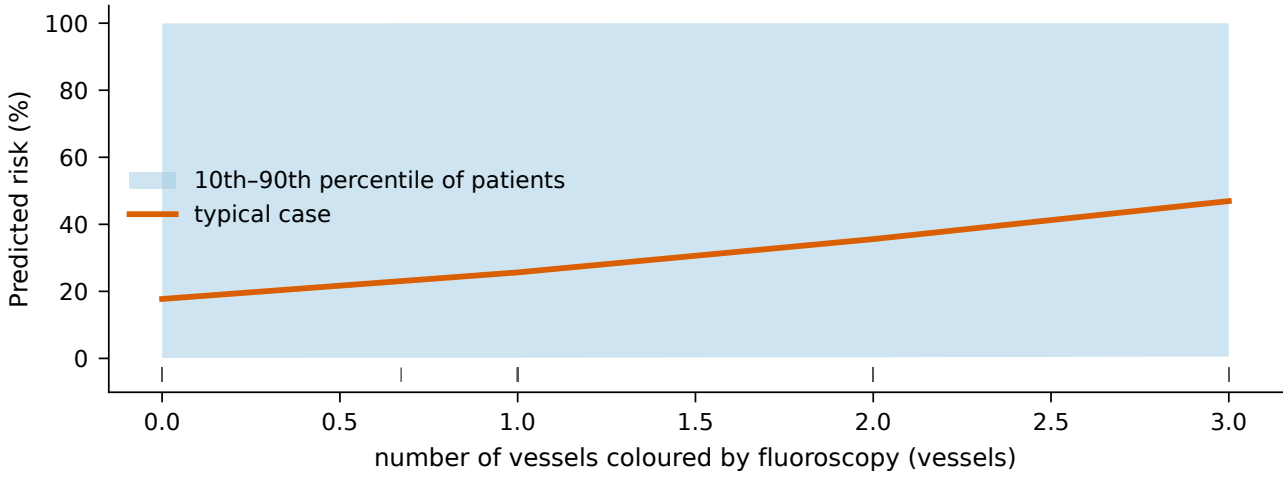


Figure 23: Vessels with flat ST segment: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

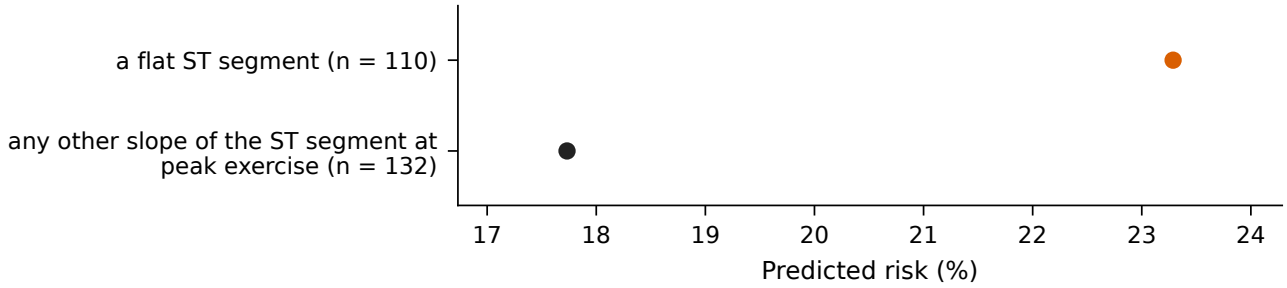


Figure 24: Vessels with flat ST segment: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

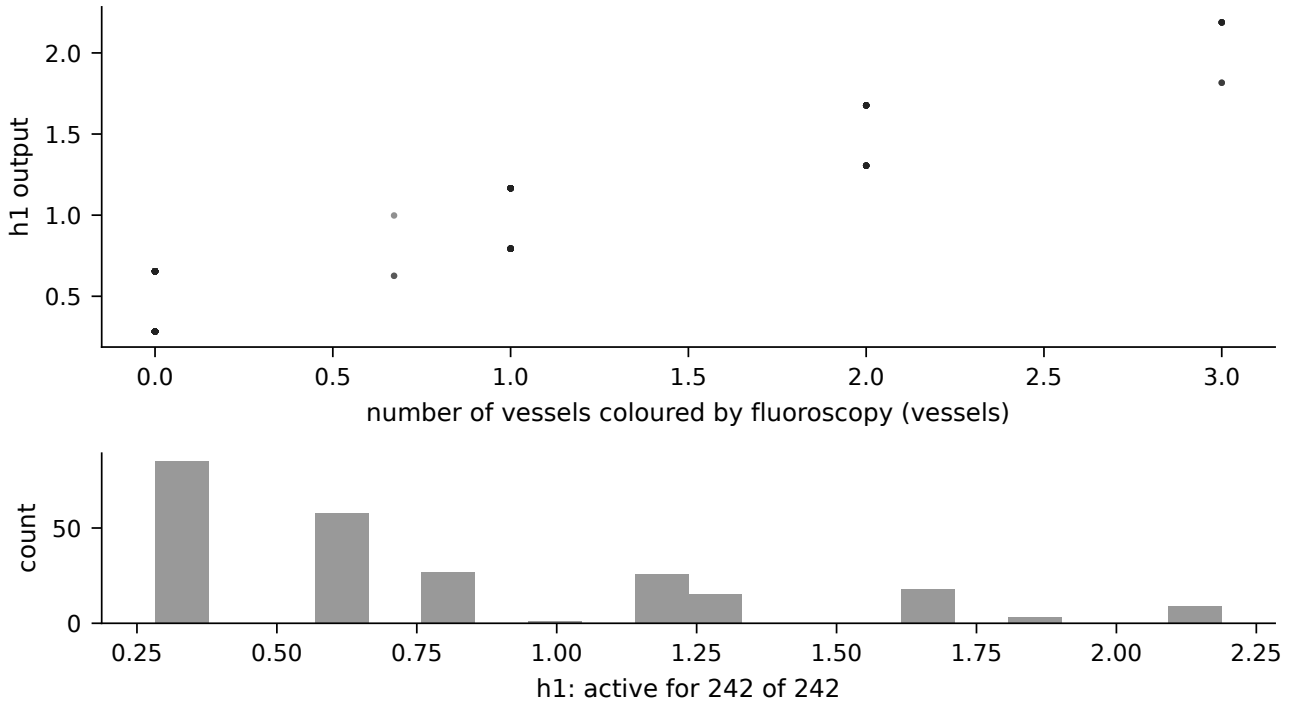


Figure 25: Vessels with flat ST segment: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.4.

1.2.5 Non-anginal chest pain relay

Part of: The whole model (1.2.1).
Figure 26 shows what feeds this part; Figure 27 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 26: Non-anginal chest pain relay: what feeds it, one level down.

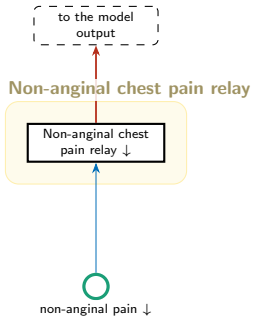


Figure 27: Non-anginal chest pain relay: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for patients with non-anginal chest pain. [C43]

Evidence. Through it, non-anginal chest pain takes a typical patient's predicted risk from about 18% to about 8%. [C43] Its formula reproduces the part exactly on all 242 training patients. [C44]
Evidence plots: Figure 28, Figure 29.

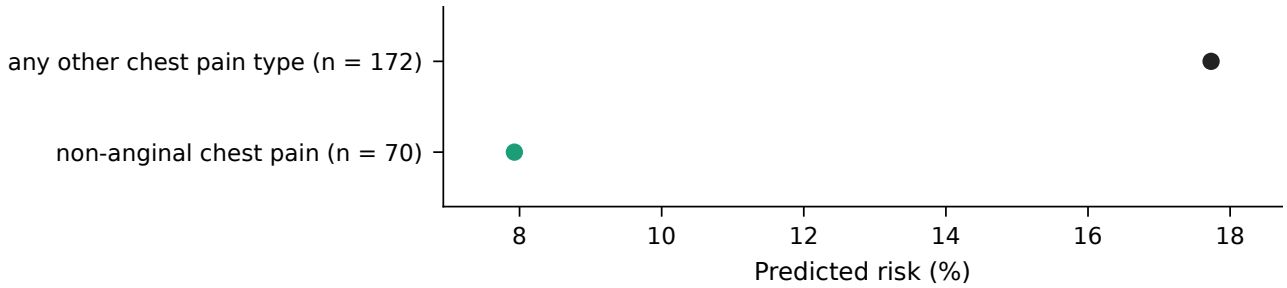


Figure 28: Non-anginal chest pain relay: The predicted value for the typical case at each level, through this part (n = training patients; grey: fewer than 20).

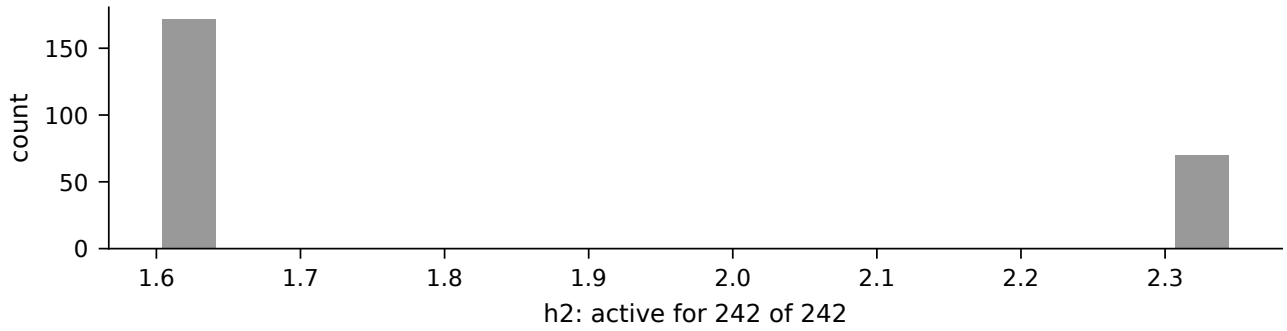


Figure 29: Non-anginal chest pain relay: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.5.

1.2.6 Maximum heart rate

Part of: The whole model (1.2.1).
Figure 30 shows what feeds this part; Figure 31 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 30: Maximum heart rate: what feeds it, one level down.

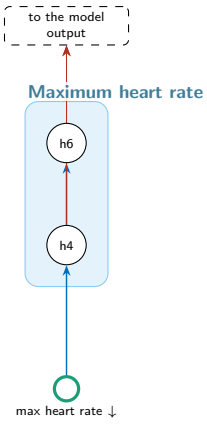


Figure 31: Maximum heart rate: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part lowers the predicted risk for a higher maximum heart rate: nothing until about 145 beats per minute, then steadily. [C45, C46]

Evidence. Through it, a higher maximum heart rate takes a typical patient from about 19% at 71 to about 13% at 182 beats per minute, with nothing until about 145 beats per minute, then steadily. [C46] Its formula reproduces the part exactly on all 242 training patients. [C47]
Evidence plots: Figure 32, Figure 33, Figure 34, Figure 35.
Figure 33 shows this part's own output, unit h4: it switches on at about 144.7 beats per minute of maximum heart rate, and is zero below.
Figure 34 shows this part's own output, unit h6.

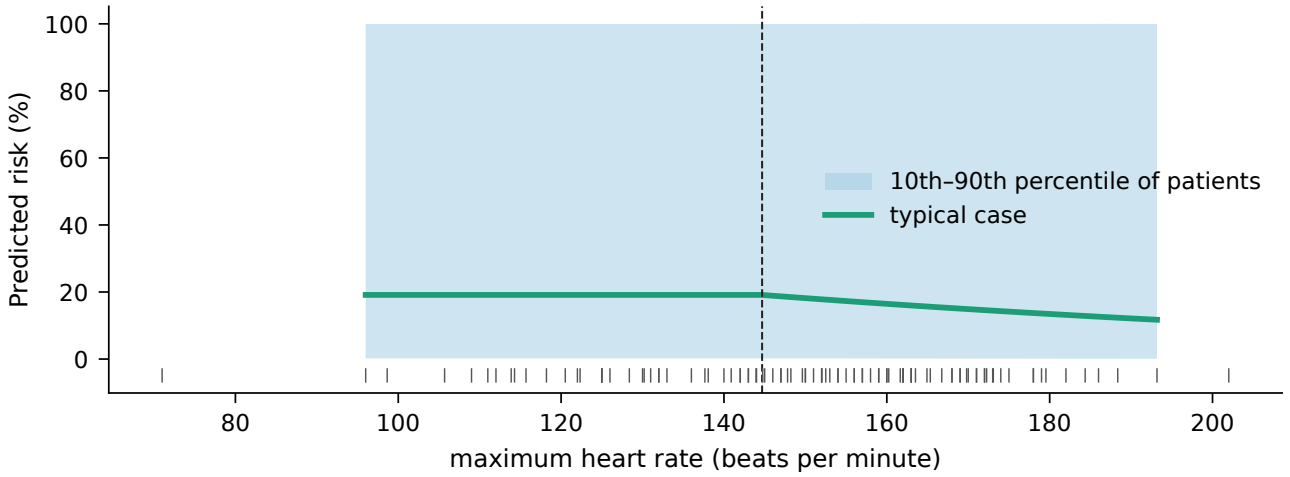


Figure 32: Maximum heart rate: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual patients (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

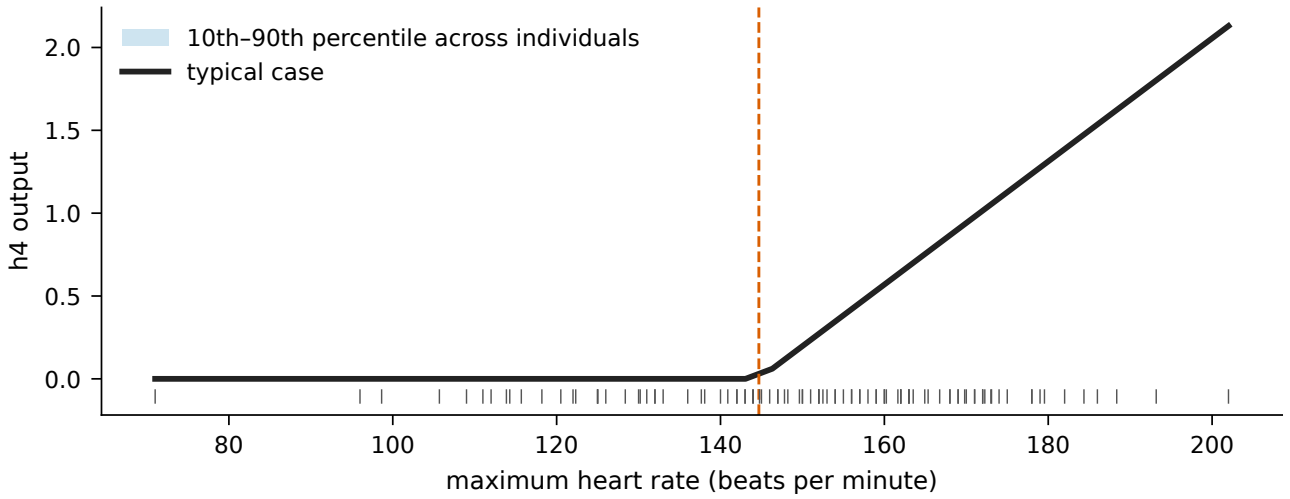


Figure 33: Maximum heart rate: This part's own output, unit h4 (the output of 'Maximum heart rate'), on its own scale (its activation), with every other input at the typical case; it is active for 155 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

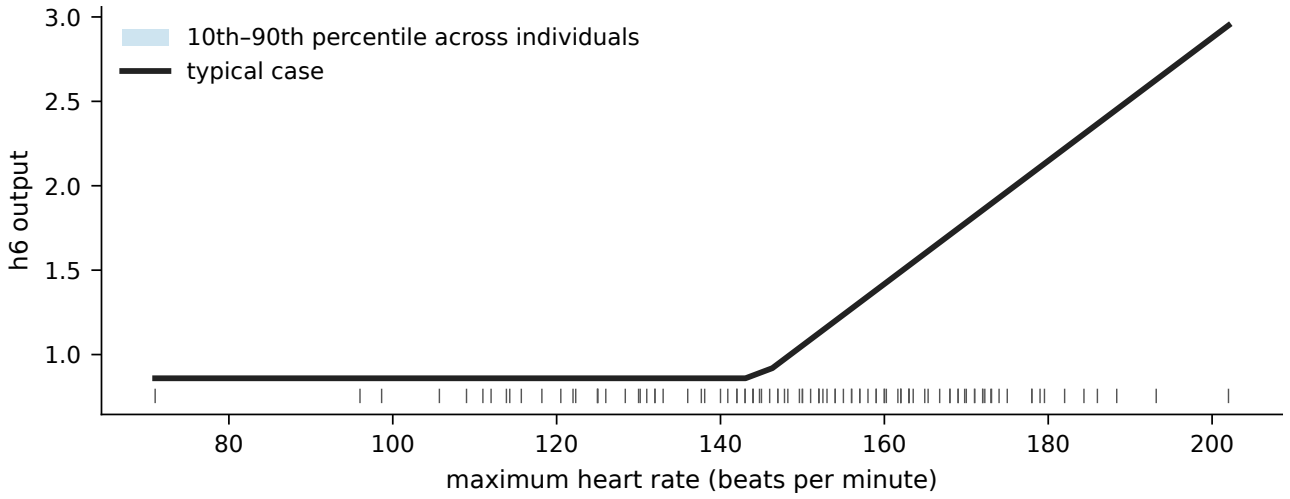


Figure 34: Maximum heart rate: This part's own output, unit h6 (the output of 'Maximum heart rate'), on its own scale (its activation), with every other input at the typical case; it is active for 242 of 242. Bold: the typical case; band: 10th to 90th percentile across individuals; ticks: the data.

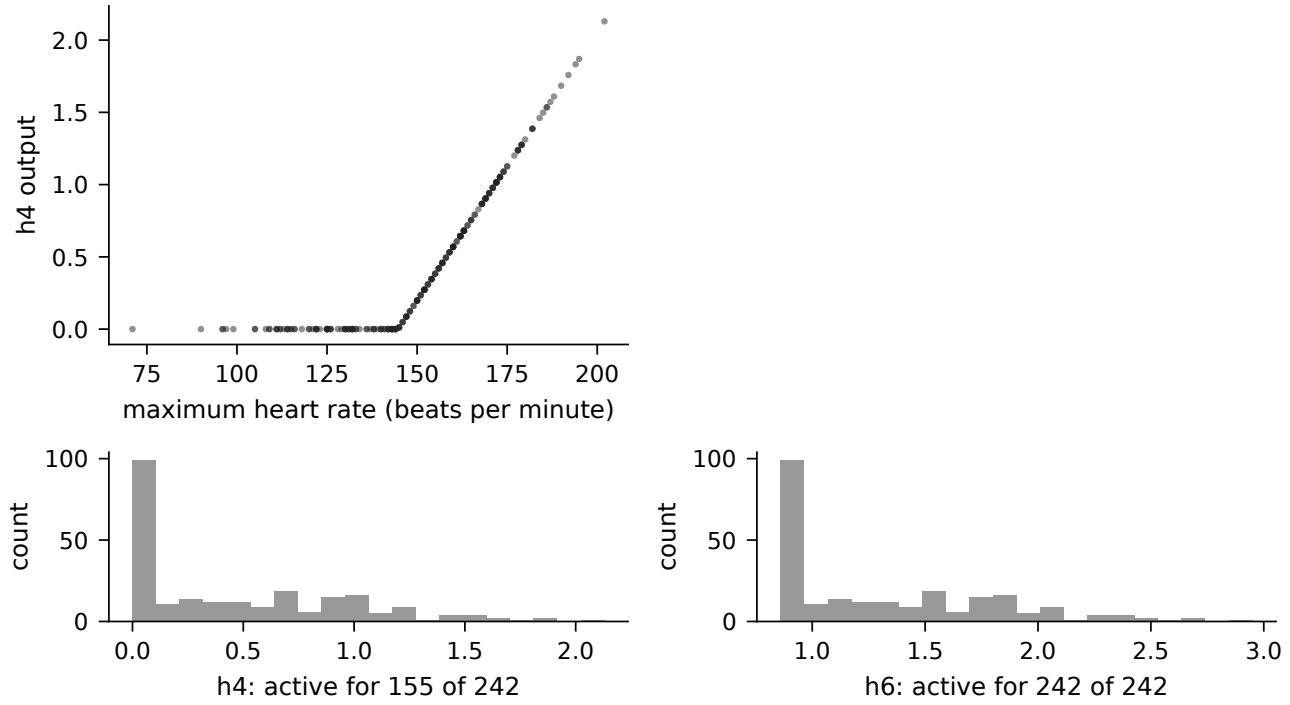


Figure 35: Maximum heart rate: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.6.

1.3 Not supported by the model

Every claim passed its checks.

Technical detail: heart-s1-c1

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_34 (heart_disease_logit). Reads: age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG, maximum heart rate reached during the exercise test, exercise-induced angina, ST depression during exercise (compared with rest), slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy, thallium stress test result.

Original hypothesis (unchanged). Hypothesis: the heart-disease logit is essentially a logistic regression. 13 direct input->output edges (largest: sex:0 -1.87, slope:2 +1.72, ca +1.30, cp:4 +1.25, chol +0.93, thal:3 -0.87, thal:6 -0.86, trestbps +0.67, cp:2 +0.44, age +0.40, fbs:1 -0.32, slope:3 +0.20, thalach +0.002) are added to five children: ca_slope2_relay and the 2245 relay (raise the effective ca weight to 2.60 per SD), cp3_relay (non-anginal pain lowers risk), st_depression_normal_ecg_gate (ST depression raises risk only with a normal resting ECG), typical_angina_ca_switch's 3210 (typical angina lowers risk, less with more vessels) and max_heart_rate_hinge (small). Nine inputs are unconnected (sex:1, fbs:0, restecg:1, restecg:2, exang:0, exang:1, slope:1, thal:4, thal:7): exercise-induced angina is ignored. Evidence (train, n=242): AUC 0.9392; all hidden units replaced by their best linear fits 0.9355 (= L2 logistic regression 0.9355); all ReLUs as identity 0.9078; all hidden mean-ablated 0.9241. The gain over linear (+0.0037) comes from the ECG/ST gate (+0.0034 restored alone), the heart-rate hinge (+0.0008) and the typical-angina switch (+0.0003); the relays contribute none. Direct edges carry logit SD 4.25 of 5.63.

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: vessels (3 copies); flat ST (2 copies); max heart rate (2 copies).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 99%	5.89	362	5.83 to 5.83	242	unclear; AUC gain 0.041 [-0.032, 0.123]	yes
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 91%	3.81	44.9	3.81 to 3.81	115, 110	disagrees; AUC gain -0.011 [-0.065, 0.036]	yes
C03	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 87%	3.46	31.9	3.46 to 3.46	115, 110	disagrees; AUC gain -0.012 [-0.059, 0.030]	yes
C04	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (direct contribution only, not through any part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 80%	2.91	18.3	2.91 to 2.91	242	disagrees; AUC gain -0.001 [-0.039, 0.032]	yes
C05	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 55%	1.74	5.71	1.74 to 1.74	135, 92	unclear; AUC gain 0.012 [-0.017, 0.044]	yes
C06	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: hinge at 1.0 mm; 100% of patients same; fallback; matches the fold at 1.0 mm*	18% to 55%	0.87	2.38	0.00 to 1.74	242	unclear; AUC gain 0.005 [-0.005, 0.022]	yes
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	8% to 45%	2.22	9.23	2.22 to 2.22	242	disagrees; AUC gain -0.022 [-0.066, 0.011]	yes
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	9% to 35%	1.65	5.20	1.65 to 1.65	242	unclear; AUC gain 0.000 [-0.022, 0.020]	yes
C09	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to <1%	-4.55	0.01	-4.78 to -4.08	108, 22	agrees; AUC gain 0.041 [0.003, 0.083]	yes
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	18% to <1%	-3.94	0.02	-3.94 to -3.94	159, 83	unclear; AUC gain 0.023 [-0.014, 0.068]	yes
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 1%	-3.43	0.03	-3.43 to -3.43	108, 70	agrees; AUC gain 0.041 [0.003, 0.083]	yes
C12	direction: chest pain type: no chest pain (asymptomatic) to chest pain type other than atypical angina, no chest pain (asymptomatic) (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, chest pain type other than atypical angina, no chest pain (asymptomatic)*	18% to 2%	-2.51	0.08	-2.51 to -2.51	108, 92	agrees; AUC gain 0.037 [0.006, 0.074]	yes
C13	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	10% to 25%	1.05	2.87	1.05 to 1.05	242	disagrees; AUC gain -0.008 [-0.018, 0.001]	yes
C14	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 3%	-1.83	0.16	-1.83 to -1.83	135, 15 (small n)	unclear; AUC gain 0.012 [-0.017, 0.044]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C15	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 32%	0.77	2.16	0.77 to 0.77	115, 17 (small n)	disagrees; AUC gain -0.012 [-0.059, 0.030]	yes
C16	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 32%	0.77	2.16	0.77 to 0.77	115, 17 (small n)	disagrees; AUC gain -0.011 [-0.065, 0.036]	yes
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, chest pain type other than atypical angina, no chest pain (asymptomatic)*	18% to 5%	-1.34	0.26	-1.34 to -1.34	108, 42	agrees; AUC gain 0.037 [0.006, 0.074]	yes
C18	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 5%	-1.34	0.26	-1.34 to -1.34	108, 42	agrees; AUC gain 0.041 [0.003, 0.083]	yes
C19	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	18% to 8%	-0.92	0.40	-0.92 to -0.92	209, 33	disagrees; AUC gain -0.001 [-0.010, 0.009]	yes
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: hinge at 145 beats per minute; 100% of patients same; verified; matches the fold at 145 beats per minute*	19% to 14%	-0.36	0.70	-0.36 to -0.36	242	unclear; AUC gain 0.002 [0.000, 0.009]	yes
C21	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: flat; 100% of patients same; verified*	18% to 18%	0.005	1.00	0.005 to 0.005	242	unclear; AUC gain 0.000 [0.000, 0.000]	yes
C22	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (whole model) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	18% to 18%	-0.46	0.63	-1.74 to 0.00	121, 121	disagrees; AUC gain -0.008 [-0.021, 0.000]	yes
C23	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 99%	5.83	340		242	unclear; AUC gain 0.041 [-0.032, 0.123]	yes
C24	magnitude: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment, typical patient *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 91%	3.81	44.9		115, 110	disagrees; AUC gain -0.011 [-0.065, 0.036]	yes
C25	magnitude: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, typical patient *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 55%	1.74	5.71		135, 92	unclear; AUC gain 0.012 [-0.017, 0.044]	yes
C26	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: hinge at 1.0 mm; 100% of patients same; fallback; matches the fold at 1.0 mm*	18% to 55%	1.74	5.68		242	unclear; AUC gain 0.005 [-0.005, 0.022]	yes
C27	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	8% to 45%	2.22	9.23		242	disagrees; AUC gain -0.022 [-0.066, 0.011]	yes
C28	null: model ignores exercise-induced angina (exact rule)		0.00			242	train only	yes

C29	structure: made of ca_slope2_relay, cp3_relay, st_depression_normal_ecg_gate, max_heart_rate_hinge, typical_angina_ca_switch and inputs age, sex, cp, trestbps, chol, fbs, thalach, slope, ca, thal; added	242	train only	yes

T.2 Typical angina with vessels

Annotation ann_33 (typical_angina_ca_switch). Reads: chest pain type, number of major vessels coloured by fluoroscopy.

Original hypothesis (unchanged). Hypothesis: (i) node 2245 = ReLU(0.578 ca + 1.276) is an always-on ca relay adding +0.88 logit per SD of ca (output weight 1.528); (ii) node 3210 = ReLU(0.901 cp:1 - 0.644*2245 + 0.420) is a typical-angina switch: it is on only for cp:1 patients and then lowers the logit by 0.833 x its activation, from about -2.26 at the lowest ca (-0.73 SD) to about -1.22 at the highest ca (2.64 SD). So typical angina lowers predicted risk, and the reduction shrinks with more narrowed vessels. Evidence (train, n=242): 2245 never clips (0.86..2.80); 3210 active for all 22 cp:1 patients and none of the 220 others. Mean ablation AUC 2245 0.9339, 3210 0.9354; linear replacement 0.9392 / 0.9391 (3210 is a cp:1 switch whose within-group response is linear in ca).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(1.28 + 0.578 x_{ca_c}\big) \\
y_1 &= \operatorname{relu}\big(0.42 + 0.901 x_{cp:1} - 0.644 \operatorname{relu}\big(1.28 \\
&\qquad + 0.578 x_{ca_c}\big)\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C30	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (through this part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 61%	2.04	7.73	1.98 to 1.98	242	disagrees; AUC gain -0.003 [-0.032, 0.019]	yes
C31	direction: chest pain type: chest pain type other than typical angina to typical angina (through this part) *shape: categorical; highest to lowest: chest pain type other than typical angina, typical angina*	18% to 2%	-2.04	0.13	-2.26 to -1.57	220, 22	unclear; AUC gain 0.001 [-0.005, 0.008]	yes
C32	interaction: effect of number of vessels coloured by fluoroscopy (0 vessels to 2 vessels) by chest pain type: typical angina minus chest pain type other than typical angina; stronger in typical angina *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	typical angina: <1% to 55%; chest pain type other than typical angina: 18% to 99%	0.70	2.00		22, 220	agrees; AUC gain 0.001 [-0.005, 0.008]	yes
C33	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

T.3 ST depression with a normal ECG

Annotation ann_31 (st_depression_normal_ecg_gate). Reads: resting ECG, ST depression during exercise (compared with rest).

Original hypothesis (unchanged). Hypothesis: node 1801 = ReLU(1.542 restecg:0 + 0.817 oldpeak - 1.521) is a gate in which ST depression (oldpeak) raises predicted risk only for patients with a normal resting ECG (restecg:0), once oldpeak is above about its mean (threshold -0.03 SD); with an abnormal resting ECG the threshold is 3.75 SD and the gate is effectively closed. Evidence (train, n=242): active for 41 of 121 normal-ECG patients and 1 of 121 others; adds 1.40 x activation = +1.14 logit per SD oldpeak when open. It is the model's main nonlinearity: mean ablation AUC 0.9339, linear replacement 0.9362, and restoring it onto the all-linear model gains +0.0034 AUC. oldpeak has no direct edge, so the model ignores ST depression for patients with an abnormal resting ECG.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$y_0 = \mathrm{relu}\big(-1.52 + 1.54 x_{restecg:0} + 0.818 x_{oldpeak}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C34	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: hinge at 1.0 mm; 100% of patients same; fallback; one point with C37 (median fold 1.0 mm, typical patient 1.0 mm, 10th-90th percentile 1.0 mm to 1.0 mm)*	18% to 55%	0.87	2.38	0.00 to 1.74	242	unclear; AUC gain 0.005 [-0.005, 0.022]	yes
C35	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	99% to 98% (typical patient this part acts on)	-0.46	0.63	-1.74 to 0.00	121, 121	disagrees; AUC gain -0.008 [-0.021, 0.000]	yes
C36	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: hinge at 1.0 mm; 100% of patients same; fallback*	resting ECG other than a normal resting ECG: 18% to 18%; a normal resting ECG: 18% to 55%	-1.74	0.18		121, 121	agrees; AUC gain -0.008 [-0.021, 0.000]	yes
C37	threshold: ST depression on exercise: fold at about 1.0 mm (starts) for patients with a normal resting ECG *shape: hinge at 1.0 mm; 100% of patients same; fallback; one point with C37 (median fold 1.0 mm, typical patient 1.0 mm, 10th-90th percentile 1.0 mm to 1.0 mm)*	13% to 63% (typical patient with a fold (patients with a normal resting ECG))	2.41	11.1	2.41 to 2.41	121, 41	unclear; AUC gain 0.005 [-0.005, 0.022]	yes
C38	subgroup: who this part acts on: 42 of 242 patients	98% to 99% (typical patient this part acts on (risk without the part, then with it))	1.08	2.95	0.19 to 2.50	42, 200	unclear; AUC gain 0.005 [-0.006, 0.025]	yes
C39	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		2e-16			242	train only	yes

T.4 Vessels with flat ST segment

Annotation ann_29 (ca_slope2_relay). Reads: slope of the ST segment at peak exercise, number of major vessels coloured by fluoroscopy.

Original hypothesis (unchanged). Hypothesis: node 664 = ReLU(0.456 ca + 0.185 slope:2 + 0.783) is an affine relay that strengthens the direct ca and flat-ST-slope (slope:2) effects. Evidence (train, n=242): pre-activation 0.28..2.19, never clips, so the part is exactly affine on the data; it adds +0.42 logit per SD of ca and +0.17 per SD of slope:2 (output weight 0.92). Mean ablation AUC 0.9392 -> 0.9375; linear replacement leaves AUC at 0.9392. No concept beyond reweighting.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$y_0 = \mathrm{relu}\big(0.783 + 0.185 x_{slope:2_b} + 0.456 x_{ca_b}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C40	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (through this part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 36%	0.94	2.56	0.94 to 0.94	242	disagrees; AUC gain -0.002 [-0.019, 0.011]	yes
C41	direction: slope of the ST segment at peak exercise: slope of the ST segment at peak exercise other than a flat ST segment to a flat ST segment (through this part) *shape: categorical; highest to lowest: a flat ST segment, slope of the ST segment at peak exercise other than a flat ST segment*	18% to 23%	0.34	1.41	0.34 to 0.34	132, 110	disagrees; AUC gain -0.003 [-0.016, 0.005]	yes
C42	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		2e-16			242	train only	yes

T.5 Non-anginal chest pain relay

Annotation ann_30 (cp3_relay). Reads: chest pain type.

Original hypothesis (unchanged). Hypothesis: node 1173 = ReLU(0.336 cp:3 + 1.818) is an always-on relay through which non-anginal chest pain (cp:3) lowers predicted risk. Evidence (train, n=242): pre-activation 1.60..2.34, never clips (entry-exit correlation 1.0); with output weight -1.24 it contributes -0.417 logit per SD of cp:3, about -0.92 logit for non-anginal pain vs other types. cp:3 has no direct edge, so this is its only route. Mean ablation AUC 0.9373; linear replacement 0.9392.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$y_0 = \mathrm{relu}\big(1.82 + 0.336 x_{cp:3}\big)$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C43	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	18% to 8%	-0.92	0.40	-0.92 to -0.92	172, 70	unclear; AUC gain 0.001 [-0.008, 0.013]	yes
C44	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

T.6 Maximum heart rate

Annotation ann_32 (max_heart_rate_hinge). Reads: maximum heart rate reached during the exercise test.

Original hypothesis (unchanged). Hypothesis: a small hinge on maximum heart rate (thalach). Node 2009 = ReLU(0.833 thalach + 0.201) feeds the output (-0.151) and the always-on relay 2650 = ReLU(0.982*2009 + 0.859) (-0.172), a net -0.32 x 2009: higher maximum heart rate lowers predicted risk, but only above about -0.24 SD; below that it has no effect (the direct thalach edge, +0.002, is negligible). Evidence (train, n=242): 2009 active for 64% of patients; logit contribution SD 0.16 (small); mean ablation AUC 0.9391, linear replacement 0.9392, restore-onto-linear gain +0.0008.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 0.0).

$$\begin{aligned} y_0 &= \mathrm{relu}\big(0.201 + 0.833 x_{thalach_b}\big) \\ y_1 &= \mathrm{relu}\big(0.859 + 0.982\,\mathrm{relu}\big(0.201 \\ &\qquad + 0.833 x_{thalach_b}\big)\big) \end{aligned}$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C45	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 145 beats per minute; 100% of patients same; verified; one point with C46 (median fold 145 beats per minute, typical patient 145 beats per minute, 10th-90th percentile 145 beats per minute to 145 beats per minute)*	19% to 14%	-0.36	0.70	-0.36 to -0.36	242	unclear; AUC gain 0.002 [0.000, 0.009]	yes
C46	threshold: maximum heart rate: fold at about 145 beats per minute (starts) *shape: hinge at 145 beats per minute; 100% of patients same; verified; one point with C46 (median fold 145 beats per minute, typical patient 145 beats per minute, 10th-90th percentile 145 beats per minute to 145 beats per minute)*	19% to 13% (typical patient with a fold)	-0.44	0.64	-0.44 to -0.44	242, 155	unclear; AUC gain 0.002 [0.000, 0.009]	yes
C47	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		0.00			242	train only	yes

The network before any explanation

Figure 36 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

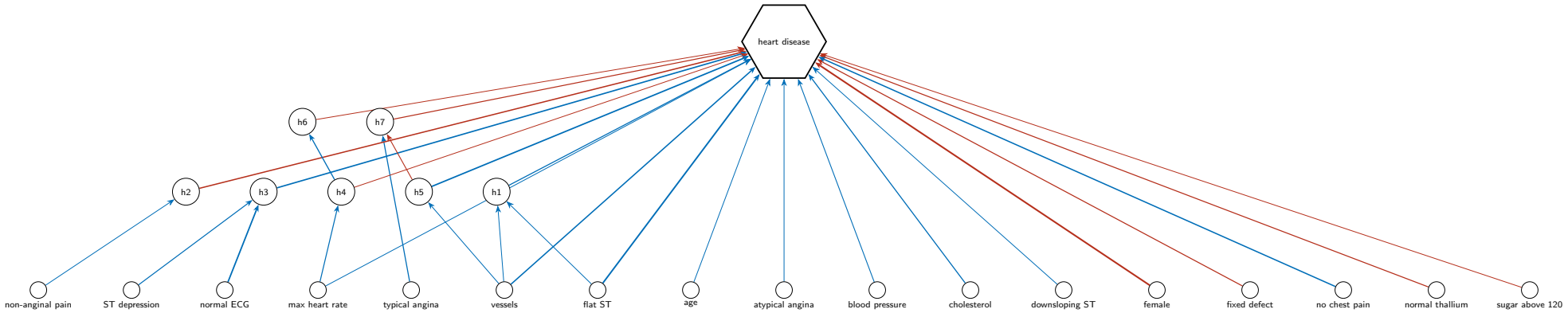


Figure 36: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): male, sugar 120 or less, ST-T abnormal ECG, LVH on ECG, no exercise pain, exercise pain, upsloping ST, thallium code 4, reversible defect.